

HDOC - Day 39

Variadic Function and Command-Line Arguments

Question 1 - Weighted Average

Algorithm: Validate count. Start a va_list using va_start. For each value-weight pair, read both as double using va_arg, reject a negative weight, accumulate value*weight and total weight, then call va_end. If total weight is zero display an error; otherwise return weighted sum / total weight.

Test Cases

Case	Input	Expected
1	80(.3),75(.2),90(.5)	84.00
2	Negative weight	Error
3	All weights zero	Error

C Program - weighted.c

```
#include <stdio.h>
#include <stdarg.h>

double weightedAverage(int count, ...) {
    va_list args; int i; double value, weight, sum=0, totalWeight=0;
    if(count<=0){printf("Invalid count.\n");return 0;}
    va_start(args, count);
    for(i=0; i<count; i++){
        value=va_arg(args, double);
        weight=va_arg(args, double);
        if(weight<0){printf("Error: weights must be non-negative.\n");va_end(args);return 0;}
        sum+=value*weight; totalWeight+=weight;
    }
    va_end(args);
    if(totalWeight==0){printf("Error: total weight cannot be zero.\n");return 0;}
    return sum/totalWeight;
}

int main(){
    double ans=weightedAverage(3,80.0,0.3,75.0,0.2,90.0,0.5);
    printf("Weighted average = %.2f\n",ans);
    return 0;
}
```

```
puppie@Anvikas-MacBook-Air Desktop % clang weighted.c -o weighted
puppie@Anvikas-MacBook-Air Desktop % ./weighted
Weighted average = 84.00
```

Question 2 - Command-Line Integer Processing

Algorithm: Read argv[1] through argv[argc-1]. Validate each complete argument as an integer and ignore invalid values. Store valid integers, sort ascending, then calculate minimum, maximum, arithmetic average, median, and scan backward for the second-largest value distinct from the maximum.

Test Cases

Case	Arguments	Expected
1	12 5 abc 8 12 20 3	3 5 8 12 12 20; min 3; max 20; avg 10; median 10; second-larg
2	abc xyz	No valid integers
3	5 5 5	Second-largest unavailable

C Program - program.c

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
int validInt(char *s){char *end;strtol(s,&end,10);return *s!='\0'&&*end=='\0';}
void sort(int a[],int n){int i,j,t;for(i=0;i<n-1;i++)for(j=0;j<n-1-i;j++)if(a[j]>a[j+1]){t=a[j];a[j]=a[j+1];a[j+1]=t;}}
int main(int argc,char *argv[]){
    int a[argc],n=0,i;double sum=0,median;
    for(i=1;i<argc;i++){if(validInt(argv[i]))a[n++]=atoi(argv[i]);else printf("Invalid argument ignored: %s\n",argv[i]);}
    if(n==0){printf("No valid integers.\n");return 0;} sort(a,n);
    printf("Valid integers in ascending order: ");for(i=0;i<n;i++){printf("%d ",a[i]);sum+=a[i];}printf("\n");
    printf("Minimum: %d\nMaximum: %d\nAverage: %.2f\n",a[0],a[n-1],sum/n);
    median=(n%2)?a[n/2]:(a[n/2-1]+a[n/2])/2.0;printf("Median: %.2f\n",median);
    for(i=n-2;i>=0&&[i]==a[n-1];i--);
    if(i>=0)printf("Second-largest distinct value: %d\n",a[i]);else printf("Second-largest distinct value not available.\n");
    return 0;
}
```

Compilation & Execution

```
puppie@Anvikas-MacBook-Air Desktop % clang program.c -o program
puppie@Anvikas-MacBook-Air Desktop % ./program 12 5 abc 8 12 20 3
Invalid argument ignored: abc
Valid integers in ascending order: 3 5 8 12 12 20
Minimum: 3
Maximum: 20
Average: 10.00
Median: 10.00
Second-largest distinct value: 12
```

Result: Both programs were compiled and tested successfully for the prepared cases.